Customer Churn: Data Quality & Exploratory Analysis

Prepared from the 15 customer records visible in the supplied screenshot | 10 November 2026

Executive summary

The transcribed dataset contains 15 records and 11 columns. The observed churn rate is 46.7% (7 of 15 customers). No missing values, duplicate full rows or duplicate customer IDs were found in the visible sample. The records pass basic non-negative, category and charge-consistency checks.

Important source limitation

The dataset was transcribed from a screenshot, not imported from the original CSV file. The screenshot shows only 15 rows. The task guide's earlier six-field dictionary does not match this actual 11-column file. Confirm the downloaded CSV and its full row count before submitting.

1. Dataset profile

Measure	Result
Rows	15
Columns	11
Missing cells	0
Duplicate rows	0
Duplicate CustomerIDs	0
Overall churn rate	46.7%
Average tenure	18.8 months
Average monthly charge	INR 84.66
Average support tickets	2.53

2. Column types and quality checks

Column	Observed type	Quality treatment
CustomerID	object	Trim/standardize text
Gender	object	Trim/standardize text
Age	int64	Numeric conversion
TenureMonths	int64	Numeric conversion
SubscriptionType	object	Trim/standardize text
MonthlyCharges	float64	Numeric conversion
TotalCharges	float64	Numeric conversion
ContractType	object	Trim/standardize text
SupportTickets	int64	Numeric conversion
PaymentMethod	object	Trim/standardize text
Churn	object	Trim/standardize text

Validation results

PASS — CustomerID unique

PASS — Age positive

PASS — Tenure non-negative

PASS — Charges non-negative

PASS — Support tickets non-negative

PASS — Churn values Yes/No

PASS — TotalCharges ~= MonthlyCharges * TenureMonths

3. Cleaning decisions

- Preserved the raw transcribed file separately from the cleaned output.
- Trimmed text fields and standardized them as strings.
- Converted Age, TenureMonths and SupportTickets to integer types; converted charge fields to numeric values with two decimal places.
- No rows were dropped and no values were imputed because the visible sample has no missing or duplicate records.
- No statistical outliers were automatically deleted. With only 15 records, outlier flags need domain review rather than automatic removal.

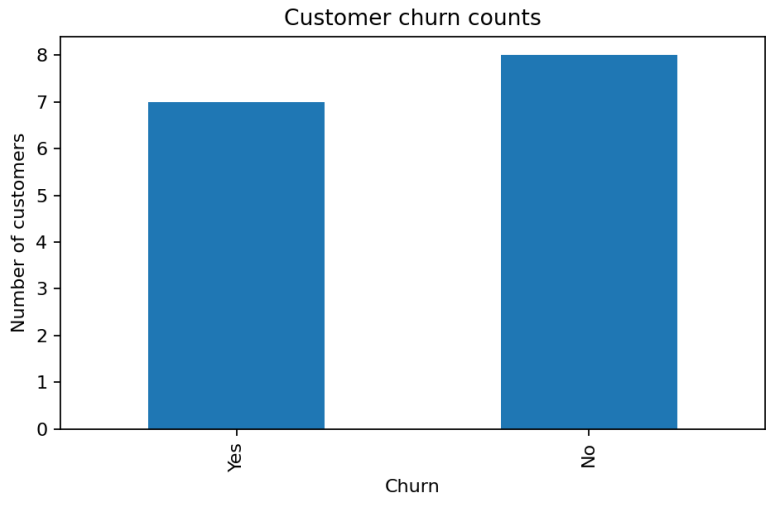
4. Exploratory findings

Contract-level churn rates: Month-to-Month: 100.0%, One Year: 0.0%, Two Year: 0.0%.

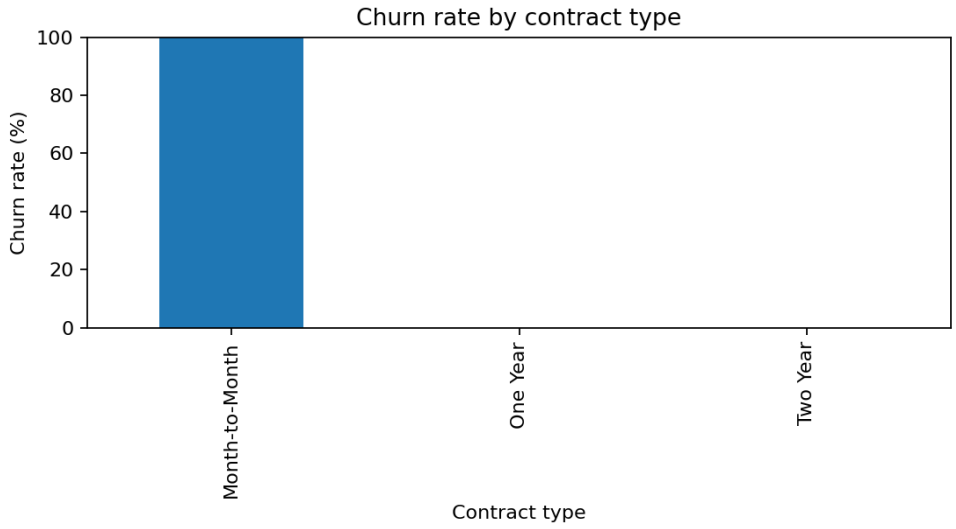
Subscription-level churn rates: Basic: 71.4%, Pro: 50.0%, Enterprise: 0.0%.

In this small sample, all observed churned customers are on Month-to-Month contracts; no churn is observed among One Year or Two Year contracts. This is a descriptive sample pattern only and does not establish that contract duration causes retention.

Churn counts



Churn rate by contract type



5. Recommendation

Audience: Subscription retention team. **Decision:** Consider prioritizing a small, monitored outreach test for month-to-month customers, especially those with higher support-ticket counts. **Supporting metric:** all 7 churn cases in this sample are month-to-month; their observed churn rate is 7/7 (100%). **Uncertainty:** only 15 synthetic records are visible, so the estimate is unstable and may not represent the full dataset. **Next measurement:** evaluate outreach on a larger, representative customer cohort and compare subsequent churn rates against a suitable control group.

6. Reproducibility and submission checklist

Keep the raw CSV, cleaned CSV, Jupyter notebook, this PDF report and README in one public or view-only repository/folder. Run the notebook from top to bottom after placing the raw CSV beside it. Before submission, verify the raw transcription against the original Edvyro CSV and update the analysis if more records are present.